CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 3: Shares and Dividend

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Return on investment

1. A company pays a 12% dividend on its shares. An investor's return on his investment is 10%. The shares were bought: (a) at par (b) below par (c) above par (d) cannot be determined ICSE 2025, 1 mark

SOLUTION

The return (10%) is less than the dividend rate (12%), so each share cost more than its nominal value.

Answer: (c)

2. A man buys ₹200 shares of a company at a premium of 25%. His return on the investment is 5%. Find (a) the market value of a share (b) the dividend per cent declared (c) the number of shares bought if his annual dividend is ₹1000. ICSE 2025, 3 marks

SOLUTION

(a) Market value $= 200 + 25\%$ of $200 = 250$. **Market value: ₹250.**

(b) Income per share $= 5\%$ of $250 = 12.50$. Dividend per cent $= \frac{12.50}{200} \times 100$. **Dividend: 6.25%.**

(c) Number of shares $= \frac{1000}{12.50}$. **80 shares.**

3. A company has 500 shares of nominal value ₹120 and declares an annual dividend of 15%. Find (a) the total dividend paid by the company (b) the annual income of a man who holds 80 shares (c) the market value of each share if his return is 10%. ICSE 2020, 3 marks

SOLUTION

Dividend on one share $= 15\%$ of $120 = 18$.

(a) Total dividend $= 500 \times 18 = 9000$

(b) Income from 80 shares $= 80 \times 18 = 1440$

(c) Return $= \frac{18}{\text{market value}} \times 100 = 10$, so market value $= 180$.

Answer: (a) ₹9000 (b) ₹1440 (c) ₹180

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4. A man invests ₹22,500 in ₹50 shares available at a 10% discount. The company pays a 12% dividend. Find (a) the number of shares bought (b) the annual dividend (c) the rate of return on his investment, to the nearest whole number. ICSE 2018, 3 marks

SOLUTION

(a) Market value $= 50 - 5 = 45$. Number of shares $= \frac{22500}{45} = 500$.

(b) Dividend on one share $= 12\%$ of $50 = 6$. Annual dividend $= 500 \times 6 = 3000$.

(c) Return $= \frac{3000}{22500} \times 100 = 13.33\%$

Answer: (a) 500 shares (b) ₹3000 (c) about 13%

5. How much must a man invest in ₹50 shares selling at ₹60 to get an income of ₹450, if the dividend is 10%? Find his yield per cent, to the nearest whole number. ICSE 2017, 3 marks

SOLUTION

Dividend on one share $= 10\%$ of $50 = 5$. Number of shares needed $= \frac{450}{5} = 90$.

Investment $= 90 \times 60 = 5400$

Yield $= \frac{5}{60} \times 100 = 8.33\%$

Answer: ₹5400; yield about 8%

2 Shares and their values

6. Shares of nominal value ₹75 are available at a discount of 20%. How much is needed to buy 15 shares? (a) ₹60 (b) ₹90 (c) ₹1350 (d) ₹900 ICSE 2024, 1 mark

SOLUTION

Market value of one share $= 75 - 20\%$ of $75 = 60$

Cost of 15 shares $= 15 \times 60 = 900$

Answer: (d)

3 Dividend

7. A man invests ₹33,000 in ₹100 shares of a company at a premium of 10%. The company declares a dividend of 12%. Find (a) the number of shares he buys (b) his annual dividend. ICSE 2024, 3 marks

SOLUTION

(a) Market value of one share $= 100 + 10 = 110$. Number of shares $= \frac{33000}{110} = 300$.

(b) Dividend on one share $= 12\%$ of $100 = 12$. Annual dividend $= 300 \times 12 = 3600$.

Answer: (a) 300 shares (b) ₹3600

8. A man invests ₹4500 in shares of a company paying a 7.5% dividend. The ₹100 shares are available at a discount of 10%. Find (a) the number of shares he buys (b) his annual income. ICSE 2019, 3 marks

SOLUTION

(a) Market value of one share $= 100 - 10 = 90$. Number of shares $= \frac{4500}{90} = 50$.

(b) Dividend on one share $= 7.5\%$ of 100 $= 7.50$. Income $= 50 \times 7.50 = 375$.

Answer: (a) 50 shares (b) ₹375

4 Selling and reinvesting

9. A man invests ₹8500 in 10%, ₹100 shares at ₹170. He sells them when the price of each share rises by ₹30, and invests the proceeds in 12%, ₹100 shares at ₹125. Find (a) the sale proceeds (b) the number of ₹125 shares he buys (c) the change in his annual income. ICSE 2019, 4 marks

SOLUTION

Shares bought first $= \frac{8500}{170} = 50$, with income $50 \times 10 = 500$.

(a) Selling price $= 170 + 30 = 200$, so the proceeds are $50 \times 200 = 10000$.

(b) New shares $= \frac{10000}{125} = 80$

(c) New income $= 80 \times 12 = 960$. Change $= 960 - 500 = 460$.

Answer: (a) ₹10,000 (b) 80 shares (c) income increases by ₹460

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